

CSE 340: Computer Architecture

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| Name: <u>Solution</u> | ID: | Section: <u>0807</u> |
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Consider a binary number,  $W = 0.000111010 \times 2^{-98}$ . Convert  $W$  in 30-bit IEEE-754 format where the size of the fraction field is 18 bits.

|          |          |          |
|----------|----------|----------|
| (1)      | (11)     | (18)     |
| sign bit | exponent | fraction |

$$\begin{aligned}
 W &= 0.000111010 \times 2^{-98} \\
 &= 1.11010 \times 2^{-98-4} \\
 &= 1.11010 \times 2^{-102} \quad [1]
 \end{aligned}$$

a) What is the Bias of this format?

Answer:  $Bias = 2^{n-1} - 1$

$$= 2^{10} - 1 = 1024 - 1 = 1023$$

b) Find the value of the Biased exponent. (answer must be in binary) [1]

Answer:  $Biased\ Exp. = Bias + Exponent$

$$= 1023 + (-102)$$

$$= 921 = 11\ 1001\ 1001 = 011\ 1001\ 100$$

c) Find the value of the fraction field. (answer must be in binary) [1]

Answer:  $1101\ 0000\ 0000\ 0000\ 00$

d) What is the value of the sign field? Answer: 0 [1]

e) Finally, complete the 30-bit representation and convert your answer into Hex. [1]

Answer:

|   |               |                        |
|---|---------------|------------------------|
| 0 | 011 1001 1001 | 1101 0000 0000 0000 00 |
|---|---------------|------------------------|